DAY_09

(Assignment)

Problem Statement: Generic Student Result Management System

Develop a Java application for managing student academic results using Generics. The application should demonstrate the use of generic classes, generic methods, and bounded wildcards (upper and lower bounds).

Requirements

Create a Student<T> generic class where T represents the type of marks obtained by a student.

The class should contain:

- Student ID
- Student name
- A collection of marks of type T
- Methods to add marks and display student details.

Tasks

1. Generic Class

- Implement a generic Student<T> class.
- Create objects such as:
    - Student<Integer> for integer marks.
    - Student<Double> for decimal marks.
- Add and display marks for different students.

2. Generic Methods

Create a generic method:

static <T> void displayMarks(List<T> marks)

The method should display all marks irrespective of their data type.

Also create a generic method to find and return the maximum value from a collection of comparable elements.

3. Upper-Bounded Wildcard

Create a method:

static void calculateTotal(List<? extends Number> marks)

The method should accept lists containing Integer, Double, Float, etc., and calculate the total marks.

Explain why **List<Integer>** and **List<Double>** can be passed to this method.

4. Lower-Bounded Wildcard

Create a method:

static void addDefaultMarks(List<? super Integer> marks)

The method should add a set of default integer marks to the supplied collection.

Demonstrate its use with:

- List<Integer>
- List<Number>
- List<Object>

Explain why these lists are accepted.

Expected Output

- Student ID: 101
- Student Name: Rahul
- Marks: [78, 85, 91]
    - Student ID: 102
    - Student Name: Priya
    - Marks: [82.5, 88.0, 91.5]
- Integer Marks Total: 254
- Double Marks Total: 262.0
- Maximum Integer Mark: 91

Default marks added successfully.

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